Laplace Transform

$\mathcal{L}\{x(t)\} = FT\{x(t)e^{-\sigma t}\}$; ROC: $\int_{-\infty}^{\infty}|x(t)e^{-\sigma t}|dt<\infty$, $x(t) \xleftrightarrow{\mathcal{L}u} X(s)=\int_{0^-}^{\infty}x(t)e^{-st}dt$

$x(t)=\frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty}X(s)e^{st}dt$	$X(s)=\int_{-\infty}^{\infty}x(t)e^{-st}dt$	ROC
$u(t)$	$\frac{1}{s}$	$Re(s)>0$
$tu(t)$	$\frac{1}{s^2}$	$Re(s)>0$
$\delta(t-\tau),\ \tau\geq 0$	$e^{-s\tau}$	all s
$e^{-at}u(t)$	$\frac{1}{s+a}$	$Re(s)>-a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$Re(s)>-a$
$[\cos(\omega_1 t)]u(t)$	$\frac{s}{s^2+\omega_1^2}$	$Re(s)>0$
$[\sin(\omega_1 t)]u(t)$	$\frac{\omega_1}{s^2+\omega_1^2}$	$Re(s)>0$
$[e^{-at}\cos(\omega_1 t)]u(t)$	$\frac{s+a}{(s+a)^2+\omega_1^2}$	$Re(s)>-a$
$[e^{-at}\sin(\omega_1 t)]u(t)$	$\frac{\omega_1}{(s+a)^2+\omega_1^2}$	$Re(s)>-a$

$ax(t)+by(t)$	$aX(s)+bY(s)$	$aX(s)+bY(s)$	At least $R_x\cap R_y$
$x(t-\tau)$	$e^{-s\tau}X(s)$ if $x(t-\tau)u(t)=x(t-\tau)u(t-\tau)$	$e^{-s\tau}X(s)$	R_x
$e^{s_0 t}x(t)$	$X(s-s_0)$	$X(s-s_0)$	$R_x+Re(s_0)$
$x(at)$	$\frac{1}{a}X(\frac{s}{a}),a>0$	$\frac{1}{\lvert a\rvert}X(\frac{s}{a})$	$\frac{R_x}{\lvert a\rvert}$
$x(t)*y(t)$	$X(s)Y(s)$ if $x(t)=y(t)=0$ for $t<0$	$X(s)Y(s)$	At least $R_x\cap R_y$
$-tx(t)$			
$\frac{d}{dt}x(t)$	$sX(s)-x(0^-)$	$sX(s)$	At least R_x
$\int_{-\infty}^{t}x(\tau)d\tau$	$\frac{1}{s}\int_{-\infty}^{0^-}x(\tau)d\tau+\frac{X(s)}{s}$	$\frac{X(s)}{s}$	At least $R_x\cap\{Re(s)>0\}$

$\delta(t-\tau),\ \tau<0$	$e^{-s\tau}$	all s
$-u(-t)$	$\frac{1}{s}$	$Re(s)<0$
$-tu(-t)$	$\frac{1}{s^2}$	$Re(s)<0$
$-e^{-at}u(-t)$	$\frac{1}{s+a}$	$Re(s)<-a$
$-te^{-at}u(-t)$	$\frac{1}{(s+a)^2}$	$Re(s)<-a$

$X(s)=\frac{b_m\prod_{k=1}^{M}(s-c_k)}{\prod_{k=1}^{N}(s-d_k)}$ zeros ∘ poles ×

property of ROC: ① no × ② ||| ③ if $x(t)$ finite duration & ab. integrable ⇒ ROC: s

④ right-sided ⇒ $\sigma>\sigma_p$
left-sided ⇒ $\sigma<\sigma_n$
two-sided ⇒ $\sigma_p<\sigma<\sigma_n$

z-Transform

$z\{x(t)\} = DTFT\{x[n]r^{-n}\}$; ROC: $\sum_{n=-\infty}^{\infty}|x[n]|r^{-n}<\infty$

$\frac{d^n}{dt^n}x(t) \xleftrightarrow{\mathcal{L}u} s^nX(s)-\frac{d^{n-1}}{dt^{n-1}}x(t)\Big|_{t=0^-}-\cdots-s^{n-1}x(0^-)$

$x[n]=\frac{1}{2\pi j}\oint X(z)z^{n-1}dz$	$X[z]=\sum_{n=-\infty}^{\infty}x[n]z^{-n}$	ROC
$\delta[n]$	1	all z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert>1$
$\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert>\lvert\alpha\rvert$
$n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\lvert z\rvert>\lvert\alpha\rvert$
$[\cos(\Omega_1 n)]u[n]$	$\frac{1-z^{-1}\cos\Omega_1}{1-z^{-1}2\cos\Omega_1+z^{-2}}$	$\lvert z\rvert>1$
$[\sin(\Omega_1 n)]u[n]$	$\frac{z^{-1}\sin\Omega_1}{1-z^{-1}2\cos\Omega_1+z^{-2}}$	$\lvert z\rvert>1$
$[r^n\cos(\Omega_1 n)]u[n]$	$\frac{1-z^{-1}r\cos\Omega_1}{1-z^{-1}2r\cos\Omega_1+r^2z^{-2}}$	$\lvert z\rvert>r$
$[r^n\sin(\Omega_1 n)]u[n]$	$\frac{z^{-1}r\sin\Omega_1}{1-z^{-1}2r\cos\Omega_1+r^2z^{-2}}$	$\lvert z\rvert>r$

$ax[n]+by[n]$	$aX(z)+bY(z)$	$aX(z)+bY(z)$	At least $R_x\cap R_y$
$x[n-k]$	see below	$z^{-k}X(z)$	R_x, except $\lvert z\rvert=0,\infty$
$\alpha^n x[n]$	$X(\frac{z}{\alpha})$	$X(\frac{z}{\alpha})$	$\lvert\alpha\rvert R_x$
$x[-n]$	—	$X(\frac{1}{z})$	$\frac{1}{R_x}$
$x[n]*y[n]$	$X(z)Y(z)$ if $x[n]=y[n]=0$ for $n<0$	$X(z)Y(z)$	At least $R_x\cap R_y$
$nx[n]$	$-z\frac{d}{dz}X(z)$	$-z\frac{d}{dz}X(z)$	R_x, except $z=0$

$u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert<1$
$-\alpha^n u[-n-1]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert<\lvert\alpha\rvert$
$-n\alpha^n u[-n-1]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\lvert z\rvert<\lvert\alpha\rvert$

$X(z)=\frac{b_0\prod_{k=1}^{M}(1-c_kz^{-1})}{\prod_{k=1}^{N}(1-d_kz^{-1})}$ zeros ∘ poles ×

property of ROC: ① no × ② ⓪ ③ if $x[n]$ finite duration ⇒ ROC: z only

④ right-sided ⇒ $r>r_+$
left-sided ⇒ $r<r_-$
two-sided ⇒ $r_+<r<r_-$

$x[n-k] \xleftrightarrow{z_u} x[-k]+x[-k+1]z^{-1}+\cdots+x[-1]z^{-k+1}+z^{-k}X(z)$

$x[n+k] \xleftrightarrow{z_u} -x[0]z^k-x[1]z^{k-1}-\cdots-x[k-1]z+z^kX(z)\ (k>0)$

$\delta[n],\ X(z)=1$

$\delta[n+1]\ X(z)=z^{-1}$

$\delta[n-1]\ X(z)=z$

	Initial Value Th.	Final Value Th.
n_y	$\lim_{s\to\infty}sX(s)=x(0^+)$	$\lim_{s\to 0}sX(s)=\lim_{t\to\infty}x(t)$
n_z	$\lim_{z\to\infty}X(z)=x[0]$ if $x[n]=0$ for $n<0$	

⇒ # of ∘ < # of × for causal $x[n]$

LT for circuit analysis

$V_R(s)=RI_R(s)$

$V_L(s)=sLI_L(s)-Li_L(0^-)$

$I_C(s)=sCV_C(s)-Cv_C(0^-)$

About DTFS

$X(e^{j\omega})=\sum_{n=-\infty}^{\infty}x[n]e^{-j\omega n}$

$\alpha^n u[n],\ \lvert\alpha\rvert<1 \leftrightarrow \frac{1}{1-\alpha e^{-j\omega}}$

$\begin{cases}1, & \lvert n\rvert<M\\ 0, & \lvert n\rvert>M\end{cases} \leftrightarrow \frac{\sin(\Omega(2M+1)/2)}{\sin(\Omega/2)}$

$\frac{\sin(Wn)}{\pi n} \leftrightarrow \begin{cases}1, & \lvert\Omega\rvert<W\\ 0, & W<\lvert\Omega\rvert<\pi\end{cases}$

$\delta[n] \leftrightarrow 1$

$1 \leftrightarrow 2\pi\delta(\Omega)$

$\lvert\alpha\rvert<1,\ \lvert\alpha\rvert<1 \leftrightarrow \frac{1-\alpha^2}{1-2\alpha\cos\Omega+\alpha^2}$

phase response

$\arg\{H(j\omega)\}=\arg\{b_m\}+\sum\arg\{j\omega-z\}-\sum\arg\{j\omega-d\}$

$A_k(d_k)^n u[n] \xleftrightarrow{z} \frac{A_k}{1-d_kz^{-1}}$ ROC: $\lvert z\rvert>d_k$

$A_{im}\frac{(n+1)\cdots(n+m-1)}{(m-1)!}(d_k)^n u[n] \xleftrightarrow{z} \frac{A_{im}}{(1-d_kz^{-1})^m}$

For ②, causal ⇒ right-sided

stable ⇒ ROC ⊃ unit circle

For ①, causal ⇒ $h(t)=0$ for $t<0$ ⇒ right-sided

stable ⇒ $\int_{-\infty}^{\infty}|x(t)e^{-\sigma t}|dt<\infty$. ROC ⊃ $Re(s)=0$

causal & stable ⇒ poles in unit circle

causal & stable ⇒ poles in l.h.

inverse $H^{inv}=\frac{1}{H}$. zeros.

zeros in l.h / unit circle

minimum phase system

$x(t)\overset{FT}{=}\frac{1}{2\pi}\int_{-\infty}^{\infty}X(j\omega)e^{j\omega t}dt,\ X(j\omega)=\int_{-\infty}^{\infty}x(t)e^{-j\omega t}dt$

DTFS: $x[n]=\sum_{k=0}^{N-1}X[k]e^{jk\omega_0 n},\ X[k]=\frac{1}{N}\sum_{n=0}^{N-1}x[n]e^{-jk\omega_0 n}$

DTFT: $x[n]=\frac{1}{2\pi}\int_{-\pi}^{\pi}X(e^{j\omega})e^{j\omega n}d\omega;\ X(e^{j\omega})=\sum_{n=-\infty}^{\infty}x[n]e^{-j\omega n}$

$(\omega_{k+1}-\omega_k)S[k]=X[k]$

Let $G_m>1$. ⇒ Tx for.

$T(s)=\frac{Y(s)}{G(s)}=\frac{GH}{1+GH}$

closed-loop ~ if $H=0$ Tx for. if $H=$ const. $y_d=1$

$E(s)=Y_d(s)-T(s)$ let $G_m=1$, $Y=Y_d+N$

Control System

open-loop ~ $y_d \to H \to G \to y$

Stability

First-order $T(s)=\frac{B(s)}{A(s)}=\frac{b_0}{s+a_0}$

Second-order $T(s)=\frac{B(s)}{A(s)}=\frac{b_1s+b_0}{s^2+a_1s+a_0}$

⇒ stable if coefficients of $A(s)$ are positive

Higher-order $A(s)=\prod_k(s+c_k)\prod(s^2+d_is+d_0)$

⇒ not stable if coefficients are not all positive

Routh-Hurwitz Criterion

$b_i=-\frac{\begin{vmatrix}a_n & a_{n-2i}\\ a_{n-1} & a_{n-2i-1}\end{vmatrix}}{a_{n-1}}$

$c_i=-\frac{\begin{vmatrix}b_1 & b_{i+1}\\ a_{n-1} & a_{n-2i-1}\end{vmatrix}}{b_1}$

s^n	a_n	$a_{n-2}\cdots$
s^{n-1}	a_{n-1}	$a_{n-3}\cdots$
s^{n-2}	b_1	$b_2\cdots$
s^{n-3}	c_1	$c_2\cdots$
$\vdots$	$\vdots$	$\vdots$

⊛ roots of sign change = # of roots in r.h. ⊛ row of zero ⇒ Usage of stability ⊛ one zero ⇒ $A(s)$ in l.h. ⊛ # of sign change

Block Diagram

Cascade $H=\prod_{i=1}^{p}H_i$: $-H_1-H_2-\cdots-H_p-$

parallel $H=\sum_{i=1}^{p}H_i$: $-\boxed{H_1}-$ $\boxed{H_2}$

$H(z)=\frac{b_0+b_1z^{-1}+b_2z^{-2}}{1+a_1z^{-1}+a_0z^{-2}}$

$H(s)=\frac{b_2+b_1s^{-1}+b_0}{1+a_1s^{-1}+a_0}$

$X(z)$ $Y(z)$ b_0 b_1 b_2 $-a_1$ $-a_0$ z^{-1}

$X(s)$ $Y(s)$ b_2 b_1 b_0 $-a_1$ $-a_0$ $\frac{1}{s}$

feedback

input $x(t)$, error $e(t)$, output $y(t)$, Gain, feedback $h(t)$

★ compare negative adder positive

$T(s)=\frac{Y(s)}{X(s)}=\frac{G(s)}{1+G(s)H(s)}$

$S_G^T=\frac{dT/T}{dG/G}=\frac{1}{1+GH}=\frac{1}{F}$

$X(s)$ $E(s)$ $G(s)$ $Y(s)$ $H(s)$ $R(s)$

y_d y H G $N(s)$ $y=Gx+n$

effect on noise

without feedback $y=Gx+n$

with feedback $y=\frac{G}{1+GH}x+\frac{1}{H(1+GH)}n$

distortion analysis

with feedback $e=a_1y+a_2y^2 \Rightarrow \frac{dx}{dy}=(a_1+H)(1+\frac{2a_2}{a_1+H}y)$

without feedback $x=a_1y+a_2y^2 \Rightarrow \frac{dx}{dy}=a_1(1+\frac{2a_2}{a_1}y)$

Reduced distortion $\Delta=\frac{1}{F}$

nonlinear $x=a_1y$ real model $x=a_1y+a_2y^2$